



**Materia:**

Programación para Entornos Virtuales

**Unidad 3:**

Ejercicios Open CV

**Alumno:**

Luis Angel Aguillon Valdez

**Carrera:**

Entornos Virtuales y Negocios Digitales

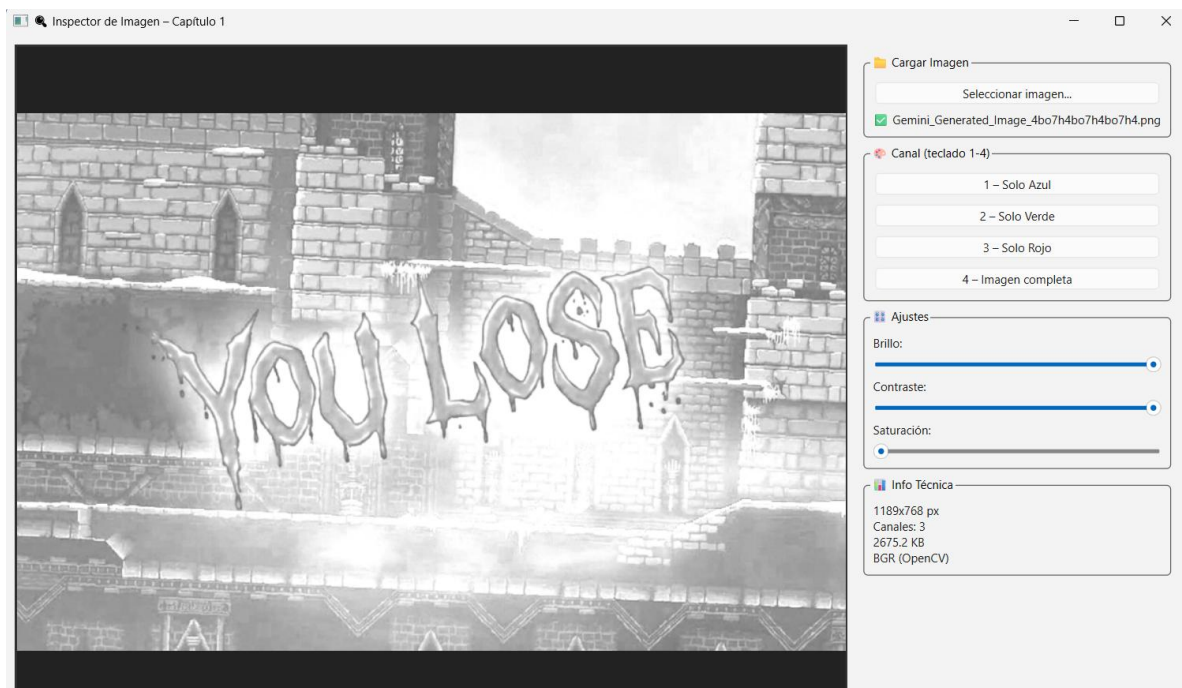
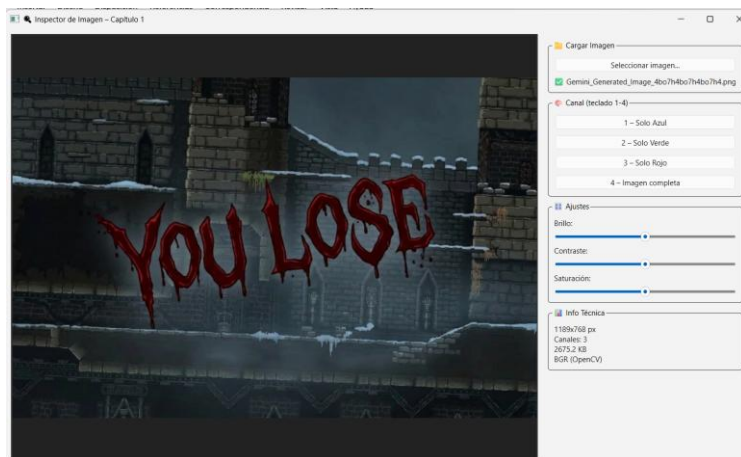
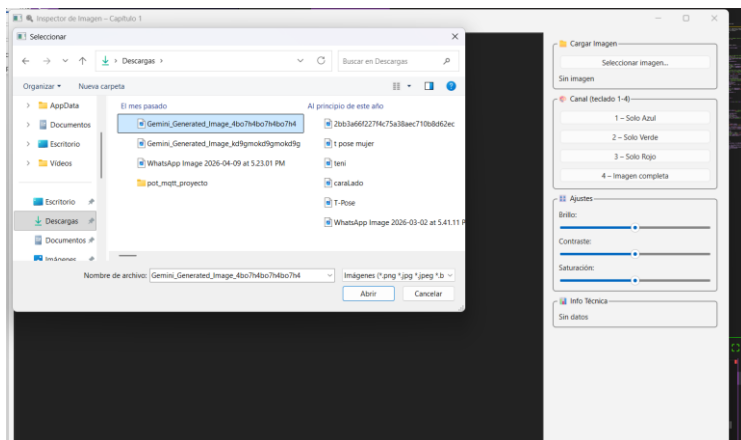
**Profesor:**

Anastacio Rodriguez Garcia

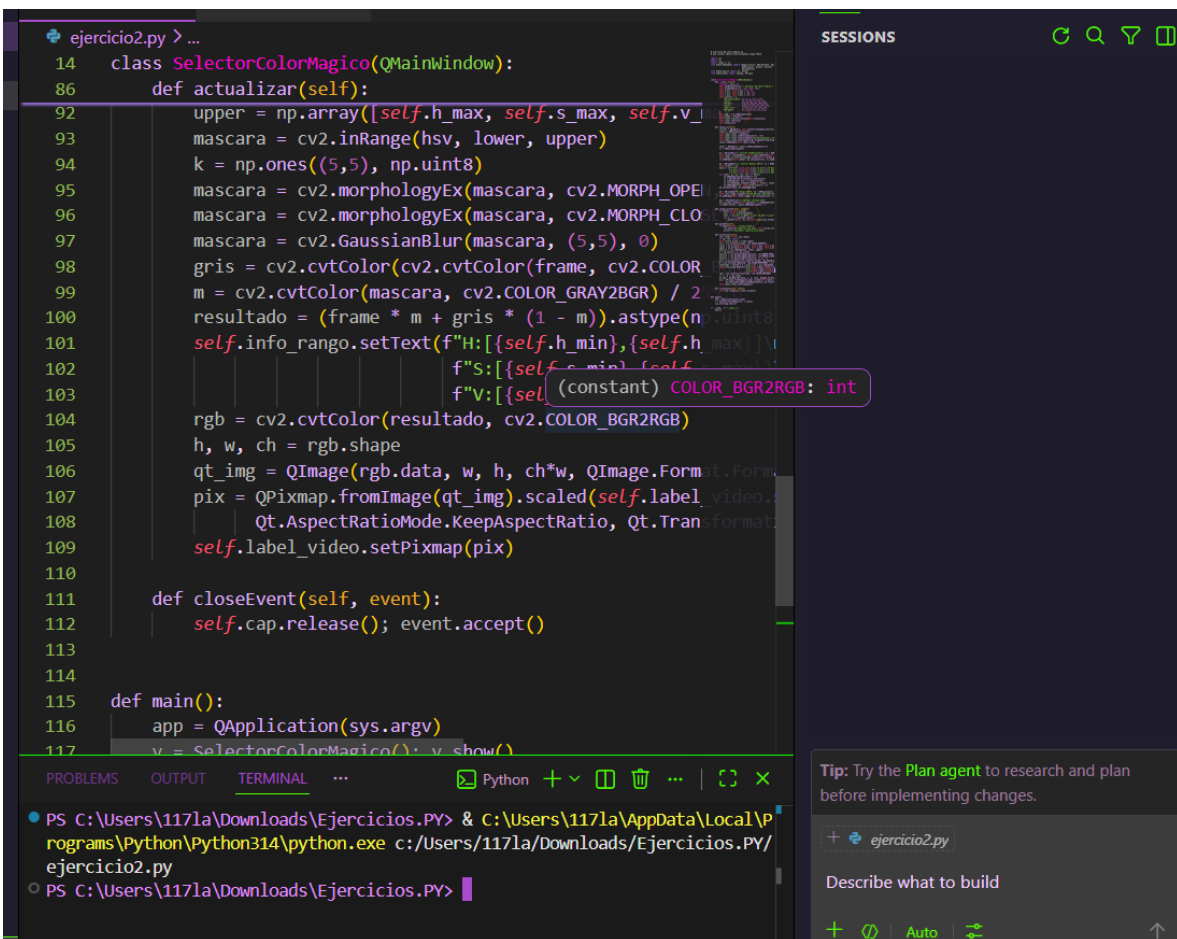
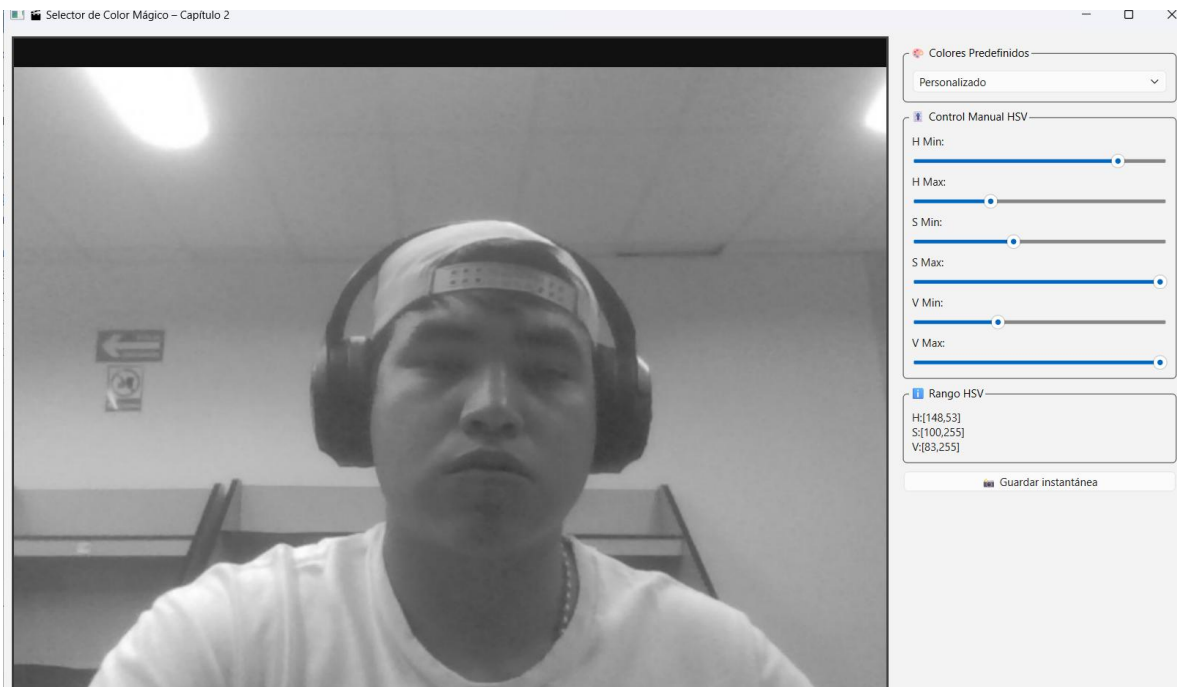
**Lugar:**

Dolores Hidalgo, Gto.

## Ejercicio 1



## Ejercicio 2



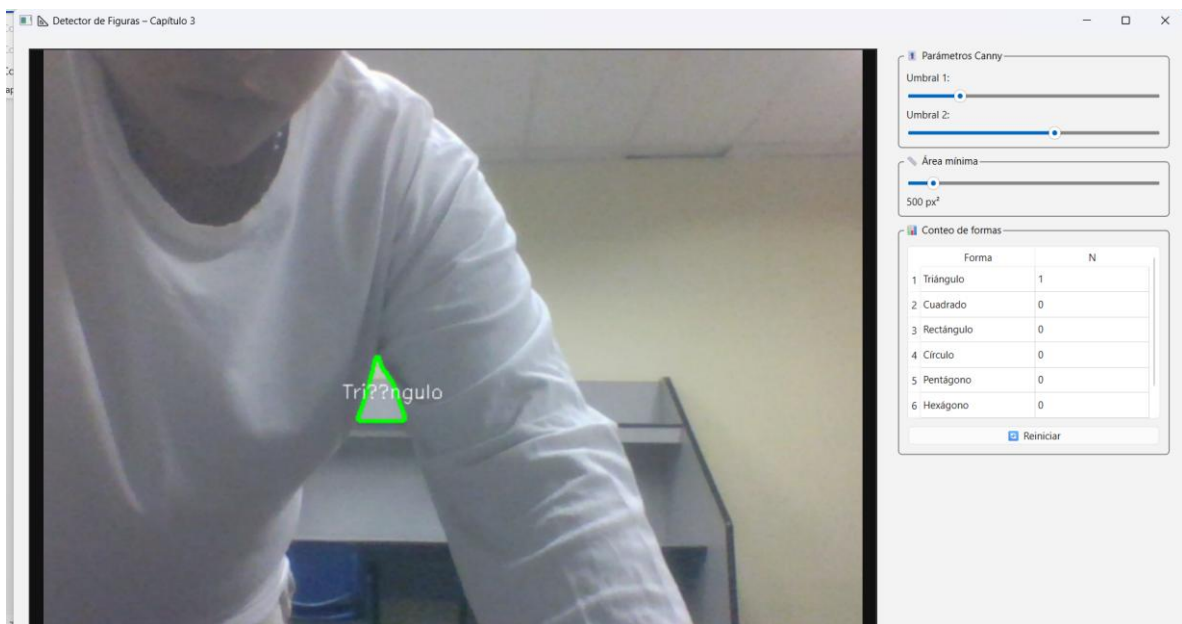
## Ejercicio 3

```

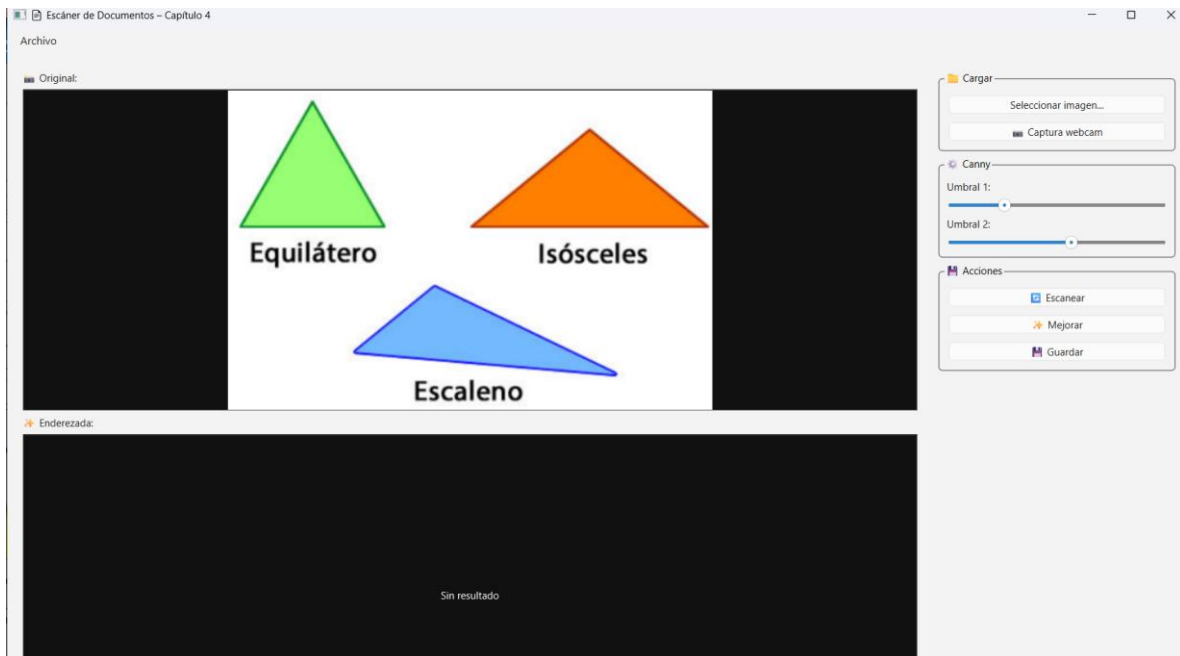
ejercicio3.py x ejercicio1.py
ejercicio3.py >
33 class DetectorFiguras(QMainWindow):
87     def actualizar(self):
98         M = cv2.moments(cnt)
99         if M["m00"] != 0:
100             cx,cy = int(M["m10"]/M["m00"]),int(M["m01"]
101             cv2.putText(frame,forma,(cx-30,cy),cv2.FO
102         for i,f in enumerate(FORMAS):
103             self.tabla.setItem(i,QTableWidgetItem(str(f
104             mostrar = cv2.cvtColor(bordes,cv2.COLOR_GRAY2BGR)
105             rgb = cv2.cvtColor(mostrar,cv2.COLOR_BGR2RGB)
106             h,w,ch = rgb.shape
107             qt_img = QImage(rgb.data,w,h,ch*4,QImage.Format_RGBA8888)
108             pix = QPixmap.fromImage(qt_img).scaled(self.label_video
109             Qt.AspectRatioMode.KeepAspectRatio,Qt.Transformation
110             self.label_video.setPixmap(pix)
111
112     def closeEvent(self, event): self.cap.release(); event.accept()
113
114 def main():
115     app = QApplication(sys.argv); v = DetectorFiguras(); v.show()
116
117 if __name__ == "__main__": main()

```

PS C:\Users\1171a\Downloads\Ejercicios.PV> C:\Users\1171a\AppData\Local\Programs\Python\Python314\python.exe c:\Users\1171a\Downloads\Ejercicios.PV\ejercicio2.py  
 PS C:\Users\1171a\Downloads\Ejercicios.PV> C:\Users\1171a\AppData\Local\Programs\Python\Python314\python.exe c:\Users\1171a\Downloads\Ejercicios.PV\ejercicio3.py  
 PS C:\Users\1171a\Downloads\Ejercicios.PV>

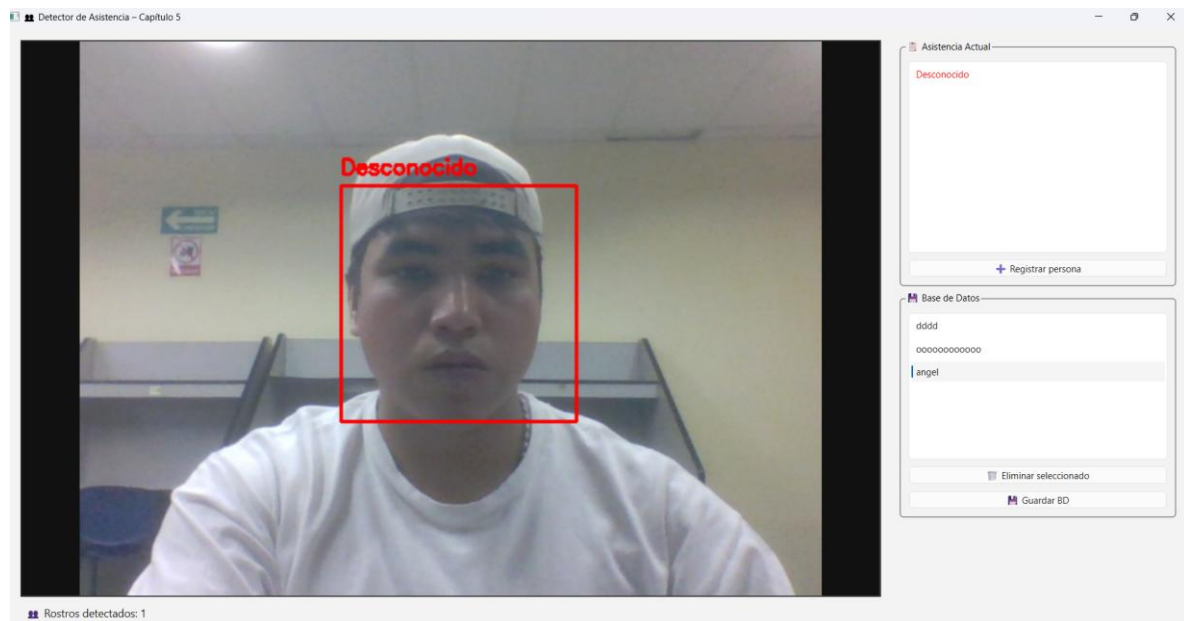


## Ejercicio 4



```
ejercicio4.py > ...
2 # pip install opencv-contrib-python numpy PyQt6
3
4 import sys
5 import cv2
6 import numpy as np
7 from datetime import datetime
8 from PyQt6.QtWidgets import (QApplication, QMainWindow, QWidget, QVBoxLayout,
9                               QHBoxLayout, QLabel, QPushButton, QGroupBox,
10                              QSlider, QFileDialog, QMessageBox)
11 from PyQt6.QtCore import Qt
12 from PyQt6.QtGui import QImage, QPixmap, QAction
13
14
15 class EscanerDocumentos(QMainWindow):
16     def __init__(self):
17         super().__init__()
18         self.setWindowTitle("Escáner de Documentos - Capítulo 4")
19         self.setGeometry(100, 100, 1400, 800)
20         self.imagen_original = None
21         self.imagen_procesada = None
22         self.setup_ui(); self.setup_menu()
23
24     def setup_ui(self):
25         central = QWidget(); self.setCentralWidget(central)
26         layout = QHBoxLayout(central)
27
28         pi = QWidget(); li = QVBoxLayout(pi)
```

## Ejercicio 5

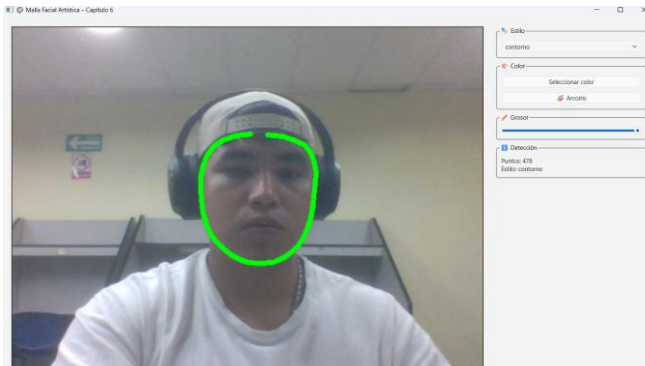


```
ejercicio2.py  ejercicio3.py  ejercicio4.py  ejercicio5.py X  ejercicio1.py  ▶ ▼ □

ejercicio5.py > ...
16  class DetectorAsistencia(QMainWindow):
    def __init__(self):
108      QMessageBox.warning(self, "Error", "No se detectó ningún rostro.")
109
110  def eliminar(self):
111      curr = self.lista_bd.currentItem()
112      if curr:
113          nombre = curr.text()
114          r = QMessageBox.question(self, "Confirmar", f"¿Eliminar a {nombre}?",
115                                  QMessageBox.StandardButton.Yes | QMessageBox.StandardButton.No)
116          if r == QMessageBox.StandardButton.Yes:
117              del self.personas_conocidas[nombre]; self.guardar_personas(); self.actualizar_lista_bd()
118
119  def actualizar_lista_bd(self):
120      self.lista_bd.clear()
121      for n in self.personas_conocidas: self.lista_bd.addItem(n)
122
123  def closeEvent(self, event): self.cap.release(); event.accept()
124
125  def main():
126      app = QApplication(sys.argv); v = DetectorAsistencia(); v.show(); sys.exit(app.exec())
127
128  if __name__ == "__main__": main()
```

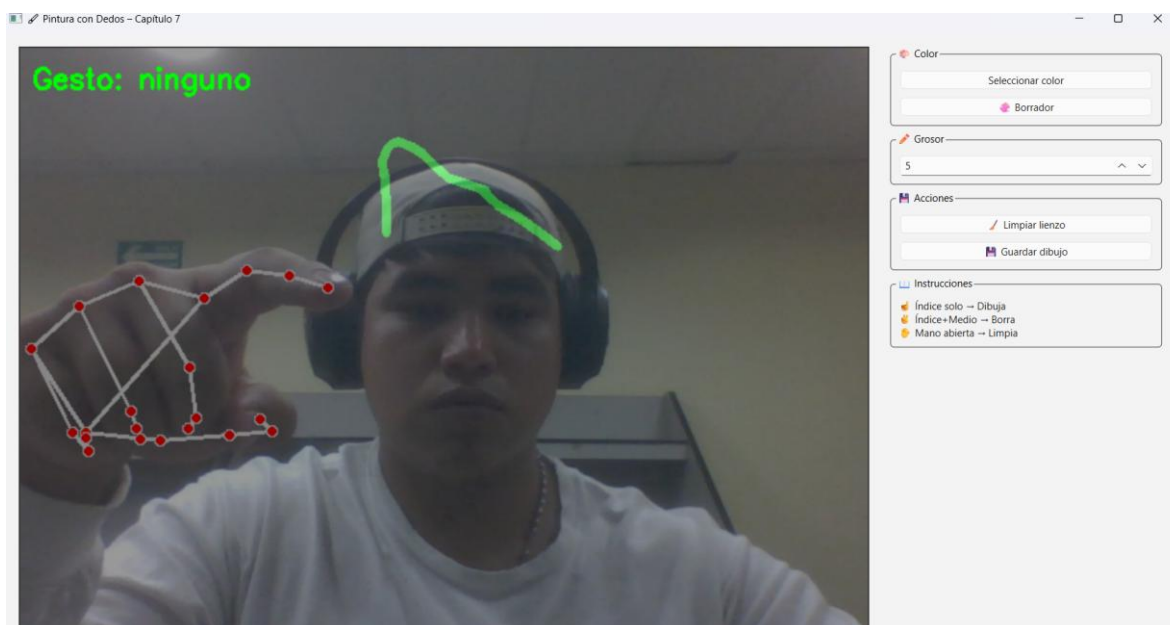
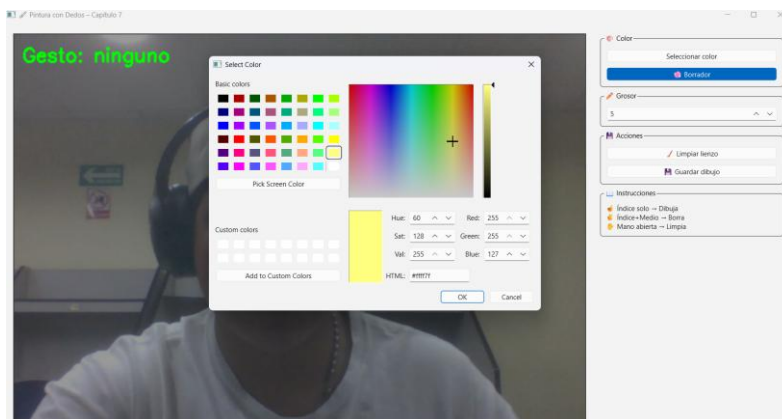


## Ejercicio 6



```
ejercicio6.py > ...
20 class MallaFacialArtistica(QMainWindow):
44 def setup_ui(self):
50     lv = QVBoxLayout(pv)
51     self.label_video = QLabel()
52     self.label_video.setMinimumSize(800, 600)
53     self.label_video.setStyleSheet("border:2px solid #333; background:#111;")
54     self.label_video.setAlignment(Qt.AlignmentFlag.AlignCenter)
55     lv.addWidget(self.label_video)
56
57     self.expr_label = QLabel("😊 Expresión: --")
58     self.expr_label.setStyleSheet("font-size:16px; font-weight:bold;")
59     lv.addWidget(self.expr_label)
60     layout.addWidget(pv, 3)
61
62     pc = QWidget()
63     pc.setMaximumWidth(350)
64     lc = QVBoxLayout(pc)
65
66     g1 = QGroupBox("🎨 Estilo")
67     v1 = QVBoxLayout()
68     self.combo = QComboBox()
69     self.combo.addItem(["contorno", "puntos", "malla_completa", "solo_ojos", "solo_boca"])
70     self.combo.currentTextChanged.connect(lambda v: setattr(self, 'estilo', v))
71     v1.addWidget(self.combo)
72     g1.setLayout(v1)
73     lc.addWidget(g1)
74
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS
PS C:\Users\1171a\Downloads\Ejercicios.PV> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:\Users\1171a\Downloads\Ejercicios.PV\ejercicio6.py
WARNING: 00:00:1779408153.310149 6084 inference_feedback_manager.cc:114] Feedback manager requires a model with a single signature inference. Disabling support for feedback tensors.
C:\Users\1171a\AppData\Local\Programs\Python\Python311\Lib\site-packages\google\protobuf\symbol_database.py:55: UserWarning: SymbolDatabase.GetPrototype() is deprecated. Please use message_factory.GetMessageClass() instead. SymbolDatabase.GetPrototype() will be removed soon.
warnings.warn('SymbolDatabase.GetPrototype() is deprecated. Please
PS C:\Users\1171a\Downloads\Ejercicios.PV>
```

## Ejercicio 7



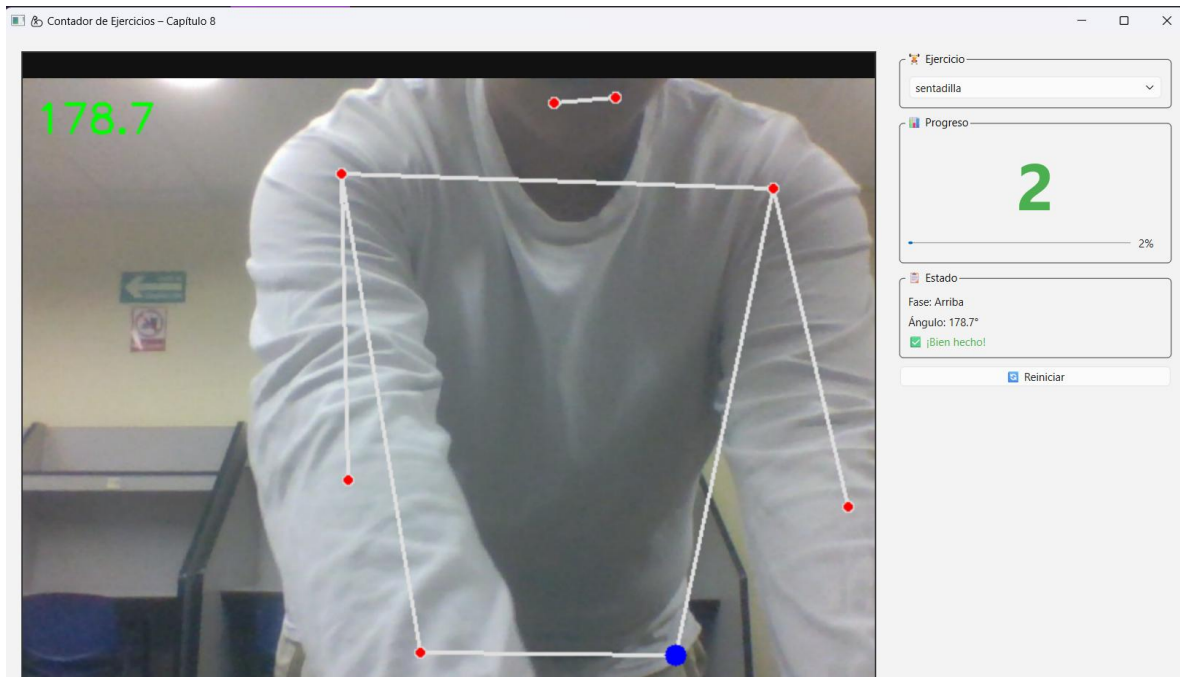
```

ejercicio7.py > ...
15 class PinturaDedos(QMainWindow):
84     def actualizar(self):
104         self.lienzo[:] = 0; self.ultima_pos = None
105         else: self.ultima_pos = None
106         mp.solutions.drawing_utils.draw_landmarks(frame, hl, self.mp_hands.HAND_CONNECTIONS)
107         else: self.ultima_pos = None
108         resultado = cv2.addWeighted(frame, 0.6, self.lienzo, 0.4, 0)
109         cv2.putText(resultado, f"Gesto: {gesto}", (10,30), cv2.FONT_HERSHEY_SIMPLEX, 0.7, self.color_actual, 2)
110         rgb2 = cv2.cvtColor(resultado, cv2.COLOR_BGR2RGB); h2,w2,ch = rgb2.shape
111         qt_img = QImage(rgb2.data,w2,h2,ch*w2,QImage.Format.Format_RGB888)
112         pix = QPixmap.fromImage(qt_img).scaled(self.label_video.size(),
113         Qt.AspectRatioMode.KeepAspectRatio,Qt.TransformationMode.SmoothTransformation)
114         self.label_video.setPixmap(pix)
115
116     def closeEvent(self, event): self.cap.release(); self.hands.close(); event.accept()
117
118     def main():
119         app = QApplication(sys.argv); v = PinturaDedos(); v.show(); sys.exit(app.exec())
120
121 if __name__ == "__main__": main()

```



## Ejercicio 8



```
ejercicio8.py > ...
12 from PyQt6.QtGui import QImage, QPixmap
13
14
15 def calcular_angulo(a, b, c, lm, w, h):
16     pa = np.array([lm[a].x*w, lm[a].y*h])
17     pb = np.array([lm[b].x*w, lm[b].y*h])
18     pc = np.array([lm[c].x*w, lm[c].y*h])
19     ba, bc = pa-pb, pc-pb
20     cos_a = np.dot(ba,bc)/(np.linalg.norm(ba)*np.linalg.norm(bc)+1e-6)
21     return float(np.degrees(np.arccos(np.clip(cos_a,-1,1))))
22
23
24 class ContadorEjercicios(QMainWindow):
25     def __init__(self):
26         super().__init__()
27         self.setWindowTitle("👤 Contador de Ejercicios 📖 Capítulo 8")
28         self.setGeometry(100, 100, 1300, 800)
29         self.mp_pose = mp.solutions.pose
30         self.pose = self.mp_pose.Pose(min_detection_confidence=0.5, min_tracking_confidence=0.5)
31         self.cap = cv2.VideoCapture(0)
32         self.timer = QTimer(); self.timer.timeout.connect(self.actualizar); self.timer.start(30)
33         self.ejercicio = "sentadilla"; self.contador = 0; self.etapa = "arriba"
34         self.historial = []
35         self.setup_ui()
36
37     def setup_ui(self):
38         central = QWidget(); self.setCentralWidget(central)
```

## Ejercicio 9

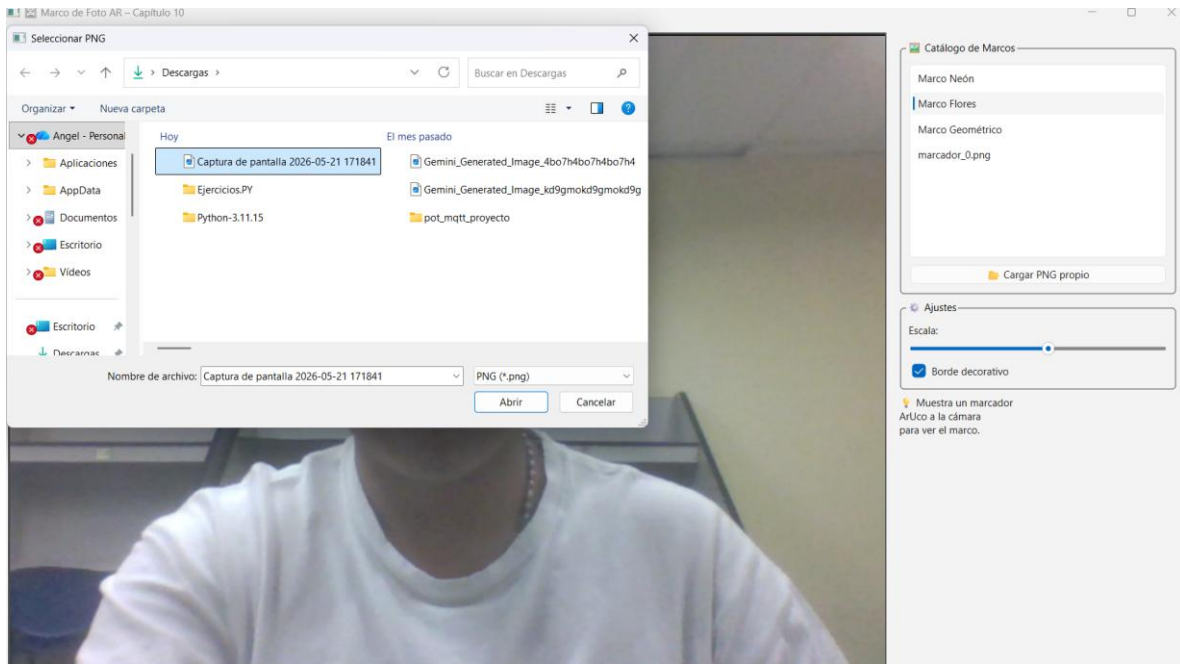


```
ejercicio9.py > ...
1 # ejercicio_09_tarjeta_ar.py
2 # pip install opencv-contrib-python numpy PyQt6
3
4 import sys
5 import cv2
6 import cv2.aruco as aruco
7 import numpy as np
8 from PyQt6.QtWidgets import (QApplication, QMainWindow, QWidget, QVBoxLayout,
9                               QHBoxLayout, QLabel, QPushButton, QGroupBox,
10                              QLineEdit, QFileDialog)
11 from PyQt6.QtCore import Qt, QTimer
12 from PyQt6.QtGui import QImage, QPixmap
13 import json
14
15
16 class TarjetaPresentacionAR(QMainWindow):
17     def __init__(self):
18         super().__init__()
19         self.setWindowTitle("Tarjeta de Presentación AR Capítulo 9")
20         self.setGeometry(100, 100, 1400, 800)
21         self.diccionario = aruco.getPredefinedDictionary(aruco.DICT_6X6_250)
22         self.detector = aruco.ArucoDetector(self.diccionario, aruco.DetectorParameters())
23         self.info_tarjetas = {
24             0: {"nombre": "Ana García", "cargo": "Ingeniera AR", "empresa": "TechVision",
25                 "email": "ana@techvision.com", "telefono": "+52 123 456 789", "color": (255, 100, 0)},
26             1: {"nombre": "Carlos López", "cargo": "Desarrollador Senior", "empresa": "AR Solutions",
27                 "email": "carlos@arsolutions.com", "telefono": "+52 987 654 321", "color": (0, 200, 100)},
28         }
29
30
31 if __name__ == '__main__':
32     app = QApplication(sys.argv)
33     window = TarjetaPresentacionAR()
34     window.show()
35     sys.exit(app.exec())
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

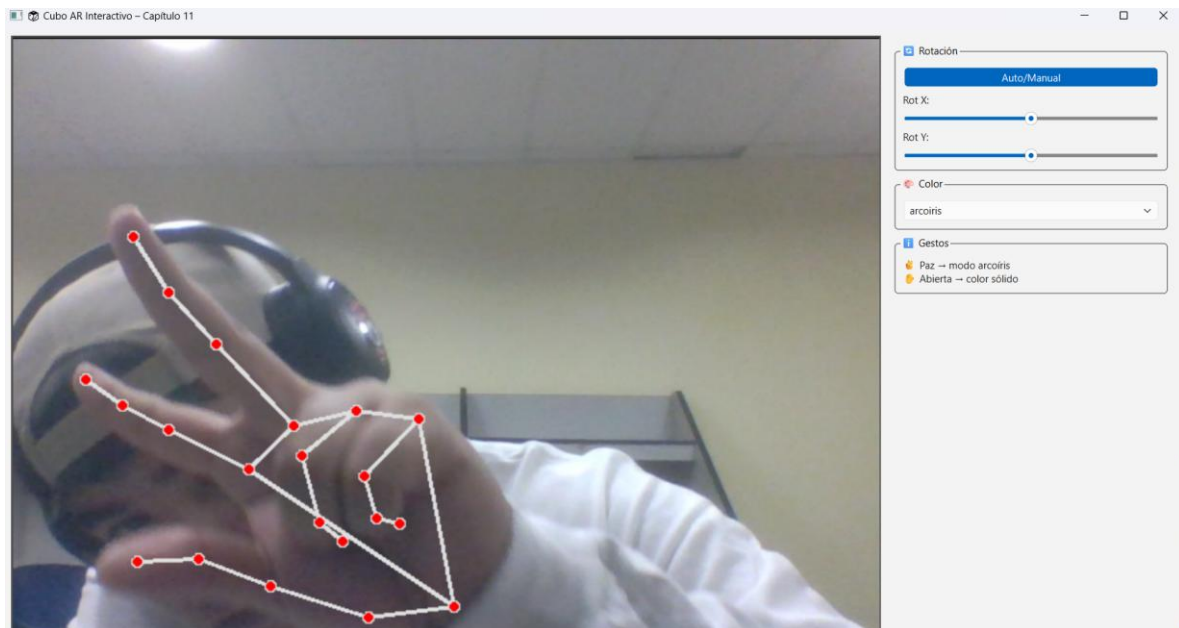
```
PS C:\Users\1171a\Downloads\Ejercicios.PY> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:/Users/1171a/Downloads/Ejercicios.PY/ejercicio9.py
Marcador 0 generado: marcador_0.png
Marcador 1 generado: marcador_1.png
Marcador 2 generado: marcador_2.png
Marcador 3 generado: marcador_3.png
PS C:\Users\1171a\Downloads\Ejercicios.PY>
```

## Ejercicio 10



```
ejercicio8.py  ejercicio9.py  ejercicio10.py X
ejercicio10.py > ...
15 def crear_marco_neon():
16     img = np.zeros((400,400,4), dtype=np.uint8)
17     for i, color in enumerate([(0,255,255,255),(255,0,255,255),(255,255,0,255)]):
18         offset = i*5
19         cv2.rectangle(img,(50+offset,50+offset),(350-offset,350-offset),color,2)
20     return img
21
22 def crear_marco_flores():
23     img = np.zeros((400,400,4), dtype=np.uint8)
24     for x,y,c in [(50,50,(255,192,203,200)),(350,50,(255,255,0,200)),(350,350,(173,216,230,200)),(50,350,(255,182,193,200))]:
25         cv2.circle(img,(x,y),40,c,-1)
26         cv2.rectangle(img,(20,20),(380,380),(255,255,255,200),3)
27     return img
28
29 def crear_marco_geometrico():
30     img = np.zeros((400,400,4), dtype=np.uint8)
31     for i in range(0,400,40):
32         for j in range(0,400,40):
33             pts = np.array([[i,j],[i+40,j],[i+20,j+20]],np.int32).reshape((-1,1,2))
34             color = (0,255,0,100) if (i+j)%80==0 else (255,0,0,100)
35             cv2.fillPoly(img,[pts],color)
36         cv2.rectangle(img,(10,10),(390,390),(255,255,255,255),5)
37     return img
38
39
40 class MarcoFotoAR(QMainWindow):
41     def __init__(self):
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## Ejercicio 11



```
ejercicio11.py > ...
26 class CuboARInteractivo(QMainWindow):
42     def setup_ui(self):
57         v1.addWidget(QLabel(lb1))
58         s1 = QSlider(Qt.Orientation.Horizontal); s1.setRange(-180,180)
59         s1.valueChanged.connect(lambda v, a=attr: setattr(self,a,v)); v1.addWidget(s1)
60         g1.setLayout(v1); lc.addWidget(g1)
61
62         g2 = QGroupBox("Color"); v2 = QVBoxLayout()
63         self.combo_color = QComboBox()
64         self.combo_color.addItem(["solido","arcoiris","por cara"])
65         self.combo_color.currentTextChanged.connect(lambda v: setattr(self,'modo_color',v))
66         v2.addWidget(self.combo_color); g2.setLayout(v2); lc.addWidget(g2)
67
68         g3 = QGroupBox("Gestos"); v3 = QVBoxLayout()
69         v3.addWidget(QLabel("Paz -> modo arcoiris\nAbierta -> color sólido"))
70         g3.setLayout(v3); lc.addWidget(g3)
71         lc.addStretch(); layout.addWidget(pc, 1)
72
73     def detectar_gesto(self, lm):
74         i = lm.landmark[8].y < lm.landmark[6].y
75         m = lm.landmark[12].y < lm.landmark[10].y
76         a = lm.landmark[16].y < lm.landmark[14].y
77         mn = lm.landmark[20].y < lm.landmark[18].y
78         if i and m and a and mn: return "abierta"
79         if i and m: return "paz"
80         return "none"
81
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\1171a\Downloads\Ejercicios.PY> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:/Users/1171a/Downloads/Ejercicios.PY/ejercicio11.py

WARNING:00:00:1779408987.229796 8120 inference\_feedback\_manager.cc:114] Feedback manager requires a model with a single signature inference. Disabling support for feedback tensors.

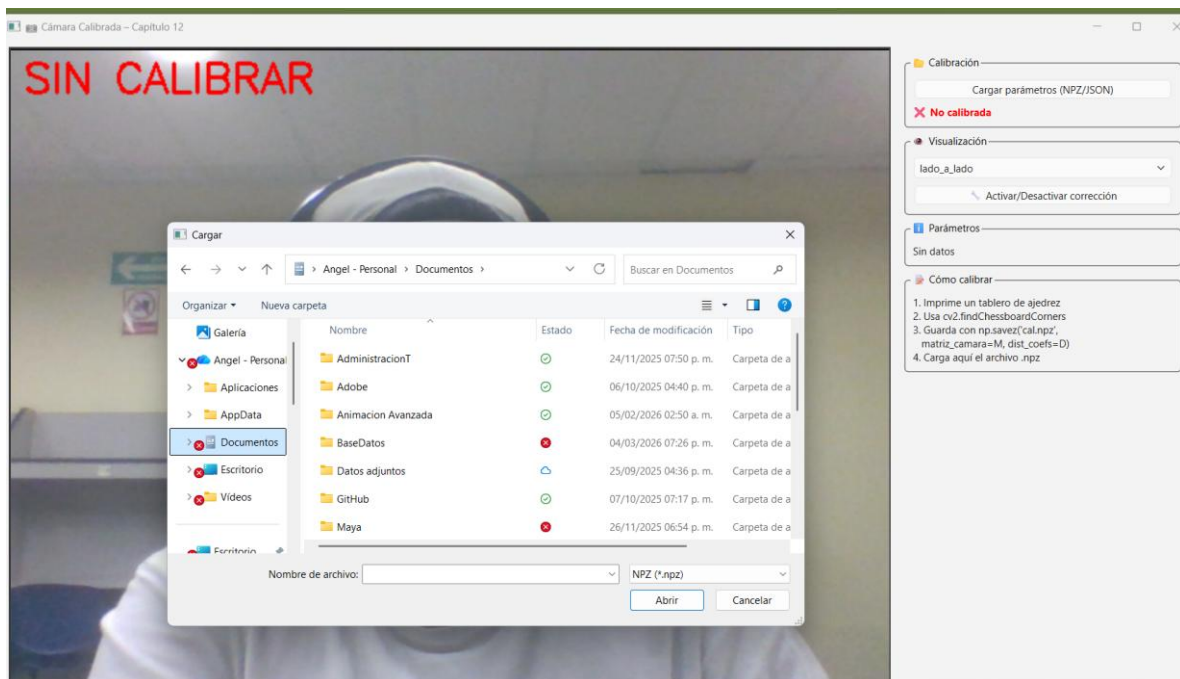
C:\Users\1171a\AppData\Local\Programs\Python\Python311\Lib\site-packages\google\protobuf\symbol\_database.py:55: UserWarning: SymbolDatabase.GetPrototype() is deprecated. Please use message\_factory.GetMessageClass() instead. SymbolDatabase.GetPrototype() will be removed soon.

warnings.warn('SymbolDatabase.GetPrototype() is deprecated. Please use message\_factory.GetMessageClass() instead. SymbolDatabase.GetPrototype() will be removed soon.')

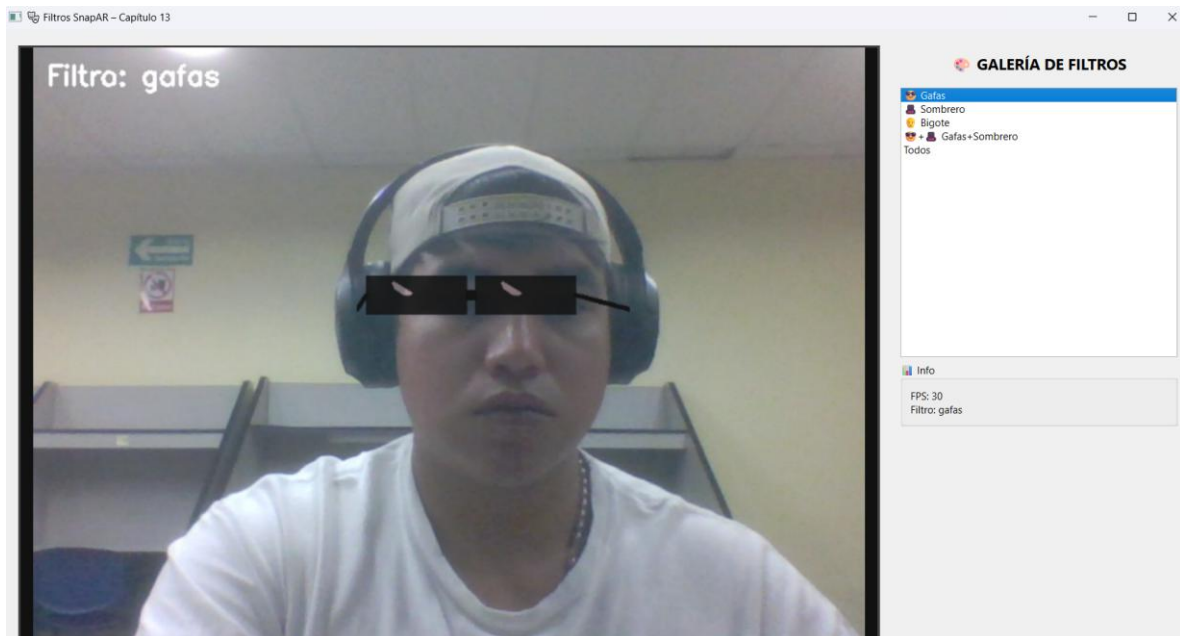
PS C:\Users\1171a\Downloads\Ejercicios.PY>

## Ejercicio 12

```
ejercicio12.py > ...
16 class CamaraCalibrada(QMainWindow):
27     def setup_ui(self):
38
39         g3 = QGroupBox("Parámetros"); v3 = QVBoxLayout()
40         self.info_text = QLabel("Sin datos"); v3.addWidget(self.info_text)
41         g3.setLayout(v3); lc.addWidget(g3)
42
43         g4 = QGroupBox("Cómo calibrar"); v4 = QVBoxLayout()
44         v4.addWidget(QLabel("1. Imprime un tablero de ajedrez\n2. Usa cv2.findChessboardCorners\n3. Guarda con np.savez('cal.npz',\n matriz_camara=M, dist_coefs=D)\n4. Carga aquí el archivo .npz"))
45         g4.setLayout(v4); lc.addWidget(g4)
46         lc.addStretch(); layout.addWidget(pc, 1)
47
48     def cargar(self):
49         f, _ = QFileDialog.getOpenFileName(self, "Cargar", "", "NPZ (*.npz);;JSON (*.json)")
50         if not f: return
51         try:
52             if f.endswith('.npz'):
53                 datos = np.load(f)
54                 self.cam_matrix = datos['matriz_camara']
55                 self.dist = datos['dist_coefs']
56             else:
57                 with open(f, 'r') as fp: datos = json.load(fp)
58                 self.cam_matrix = np.array(datos['matriz_camara'])
59                 self.dist = np.array(datos['dist_coefs'])
60             self.lbl_estado.setText("Calibrada"); self.lbl_estado.setStyleSheet("color:green; font-weight:bold;")
61
62         except:
```



## Ejercicio 13



```
ejercicio13.py > ...
59 class FiltrosSnapchat(QMainWindow):
152
153     def toggle_grabar(self):
154         if not self.grabando:
155             ts = QDateTime.currentDateTime().toString("yyyyMMdd_hhmmss")
156             nombre = f"video_{self.filtro}_{ts}.avi"
157             cap = self.cap; fw=int(cap.get(3)); fh=int(cap.get(4))
158             self.video_writer = cv2.VideoWriter(nombre, cv2.VideoWriter_fourcc(*'XVID'), 20.0, (fw,fh))
159             self.grabando = True; print(f"📹 Grabando: {nombre}")
160         else:
161             if self.video_writer: self.video_writer.release()
162             self.grabando = False; print("📹 Grabación detenida")
163
164     def closeEvent(self, event):
165         if self.grabando and self.video_writer: self.video_writer.release()
166         self.cap.release(); self.face_mesh.close(); event.accept()
167
168     def main():
169         app = QApplication(sys.argv); app.setStyle('Fusion')
170         v = FiltrosSnapchat(); v.show(); sys.exit(app.exec())
171
172 if __name__ == "__main__": main()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\1171a\Downloads\Ejercicios.PY> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:/Users/1171a/Downloads/Ejercicios.PY/ejercicio13.py

W0000 00:00:1779409322.782378 21064 inference\_feedback\_manager.cc:114] Feedback manager requires a model with a single signature inference. Disabling support for feedback tensors.

C:\Users\1171a\AppData\Local\Programs\Python\Python311\Lib\site-packages\google\protobuf\symbol\_database.py:55: UserWarning: SymbolDatabase.GetPrototype() is deprecated. Please use message\_factory.GetMessageClass() instead. SymbolDatabase.GetPrototype() will be removed soon.

warnings.warn('SymbolDatabase.GetPrototype() is deprecated. Please use ')

PS C:\Users\1171a\Downloads\Ejercicios.PY>



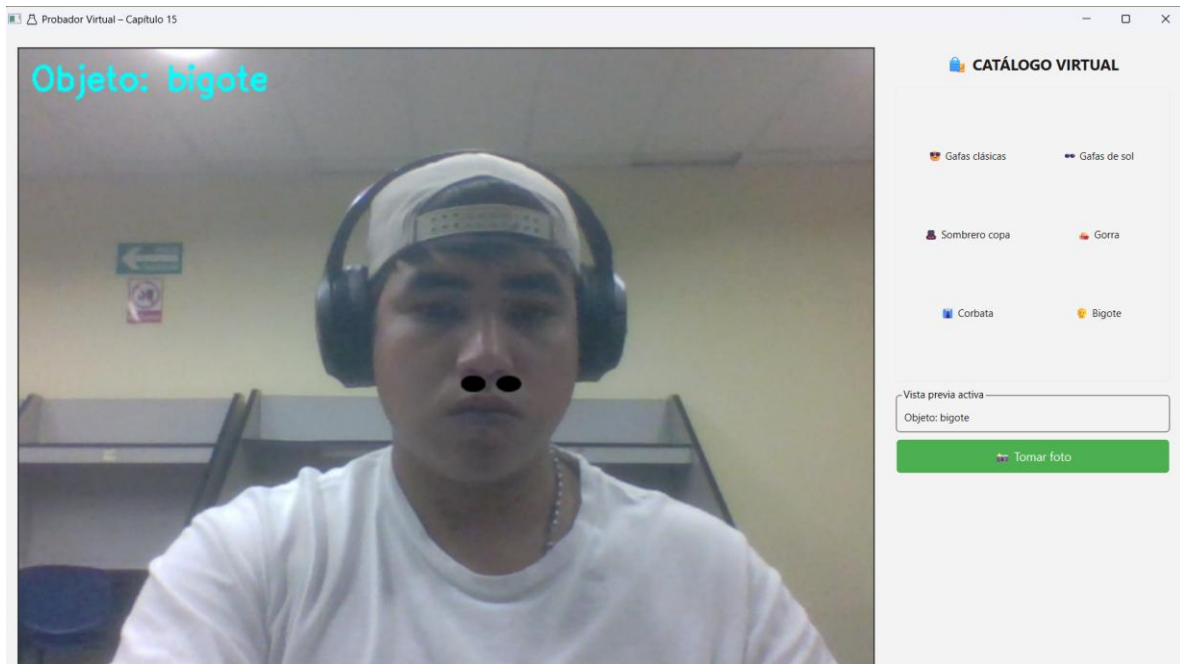
### Ejercicio 14



```
> ejercicio14.py > ...  
16 def crear_contenido(id_m):  
  
    color = colores[id_m % len(colores)]  
    img = np.zeros((200,350,4),dtype=np.uint8)  
    img[:, :, :] = color; img[:, :, 3] = 160  
  
    textos = [f"Página {str(id_m+1)}, Capítulo \"{str(id_m+1)}\", ID: \"{str(id_m)}\",  
              f\"Animales 3D\", \"Leon | Tigre | Aguila\", \"Escanea para ver mas\",  
              f\"Vehiculos AR\", \"Auto | Avion | Barco\", \"Realidad Aumentada\",  
              f\"Naturaleza\", \"Arboles | Plantas\", \"El mundo natural\"] * [id_m%4]  
  
    for i,t in enumerate(textos):  
        cv2.putText(img,t,(15,50+i*50),cv2.FONT_HERSHEY_SIMPLEX,0.65,(255,255,255),2)  
  
    return img  
  
29  
  
30 class LibroAR(QMainWindow):  
31     def __init__(self):  
32         super().__init__()  
33         self.setWindowTitle("■ Libro AR Interactivo □ Capítulo 14")  
34         self.setGeometry(100,100,1350,820)  
35         self.diccionario = aruco.getPredefinedDictionary(aruco.DICT_6x6_25) # parameter: self@LibroAR  
36         self.detector = aruco.ArucoDetector(self.diccionario, aruco.ArUcOParameters())  
37         self.contenidos = {i: crear_contenido(i) for i in range(10)}  
38         # Generar marcadores para imprimir  
39         for i in range(8):  
40             m = np.zeros((300,300),dtype=np.uint8)  
41             aruco.generateImageMarker(self.diccionario,i,300,m,1)  
42             cv2.imwrite(f"marcador_{i}.png",m)  
43         print("Marcadores 0-7 generados. Imprimelos y úsalos con el libro.")
```

```
PS C:\Users\1171a\Downloads\Ejercicios.PY> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:/Users/1171a/Downloads/Ejercicios.PY/eje  
rcio14.py  
) is deprecated. Please use message_factory.GetMessageClass() instead. SymbolDatabase.GetPrototype() will be removed soon.  
warnings.warn('SymbolDatabase.GetPrototype() is deprecated. Please '  
PS C:\Users\1171a\Downloads\Ejercicios.PY> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:/Users/1171a/Downloads/Ejercicios.PY/eje  
rcio14.py  
Marcadores 0-7 generados. Imprimelos y úsalos con el libro.
```

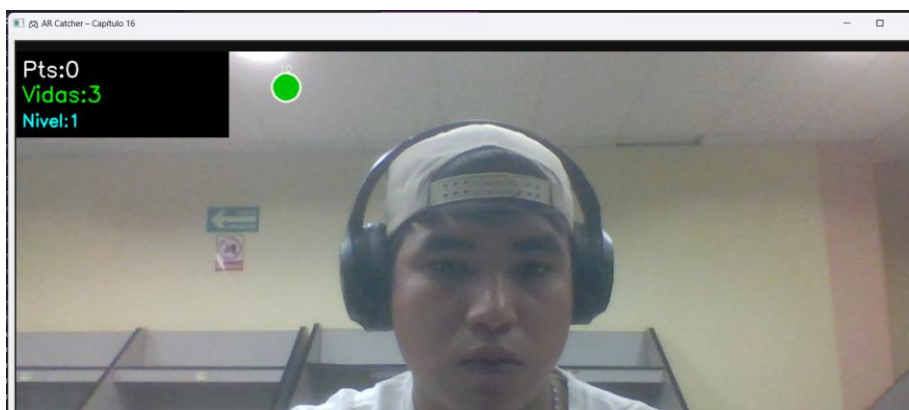
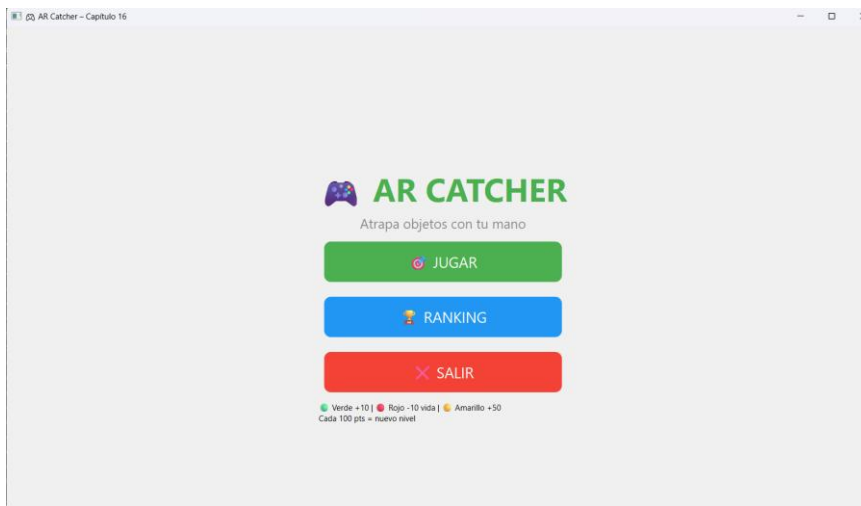
## Ejercicio 15



```
ejercicio15.py > ...
1 # ejercicio_15_probador_virtual.py
2 # pip install opencv-contrib-python numpy PyQt6 mediapipe
3
4 import sys
5 import cv2
6 import mediapipe as mp
7 import numpy as np
8 import math
9 from PyQt6.QtWidgets import (QApplication, QMainWindow, QWidget, QVBoxLayout,
10                               QHBoxLayout, QLabel, QPushButton, QGroupBox,
11                               QListWidget, QScrollArea, QGridLayout, QMessageBox)
12 from PyQt6.QtCore import Qt, QTimer
13 from PyQt6.QtGui import QImage, QPixmap
14
15 INDICES = {"ojo_izq":33,"ojo_der":362,"nariz":1,"boca_izq":61,"boca_der":291,
16           "boca_sup":0,"frente":10,"hombro_izq":11,"hombro_der":12,"menton":152}
17
18 def superponer(fondo, overlay, pos):
19     x,y = pos; h,w = overlay.shape[:2]
20     if x<0 or y<0 or x+w>fondo.shape[1] or y+h>fondo.shape[0]: return fondo
21     if overlay.shape[2]==4:
22         alpha = overlay[:,3:4]/255.0
23         roi = fondo[y:y+h,x:x+w]
24         fondo[y:y+h,x:x+w] = (roi*(1-alpha)+overlay[:,3:4]*alpha).astype(np.uint8)
25     return fondo
26
27 def mk_gafas():
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```

### Ejercicio 16



```

❏ Ejercicio16.py >
83 class ARCCatcherApp(QMainWindow):
108 def build_juego(self):
114     w = QWidget()
115     b1=QPushButton("⚙ Menú"); b1.clicked.connect(self.volver_menu); h1.addWidget(b1)
116     b2=QPushButton("🔄 Reiniciar"); b2.clicked.connect(self.reiniciar); h1.addWidget(b2)
117     lay.addWidget(h1); self.stack.addWidget(w)
118
119 def build_ranking(self):
120     w = QWidget(); lay = QVBoxLayout(w)
121     QLabel("🏆 RANKING").setParent(w); t=QLabel("🏆 RANKING"); t.setStyleSheet("font-size:32px; font-weight:bold; color:#FFD700")
122     t.setAlignment(Qt.AlignmentFlag.AlignCenter); lay.addWidget(t)
123     self.tabla_rank = QTableWidgetItem(); self.tabla_rank.setColumnCount(4)
124     self.tabla_rank.setHorizontalHeaderLabels(["Pos", "Nombre", "Puntos", "Nivel"])
125     self.tabla_rank.horizontalHeader().setSectionResizeMode(QHeaderView.ResizeMode.Stretch)
126     lay.addWidget(self.tabla_rank)
127     b=QPushButton("⬅ Volver"); b.clicked.connect(lambda: self.stack.setCurrentIndex(0)); lay.addWidget(b)
128     self.stack.addWidget(w)
129
130 def iniciar_juego(self):
131     self.motor = MotorJuego()
132     self.cap = cv2.VideoCapture(0)
133     self.cap.set(cv2.CAP_PROP_FRAME_WIDTH,1280); self.cap.set(cv2.CAP_PROP_FRAME_HEIGHT,720)
134     self.stack.setCurrentIndex(1); self.timer.start(30)
135
136 def volver_menu(self):
137     self.timer.stop()
138     if self.cap: self.cap.release(); self.cap=None
139     #if self.motor: self.motor.pause(); self.motor=None
140

```

PROBLEMS
OUTPUT
DEBUG CONSOLE
TERMINAL
PORTS

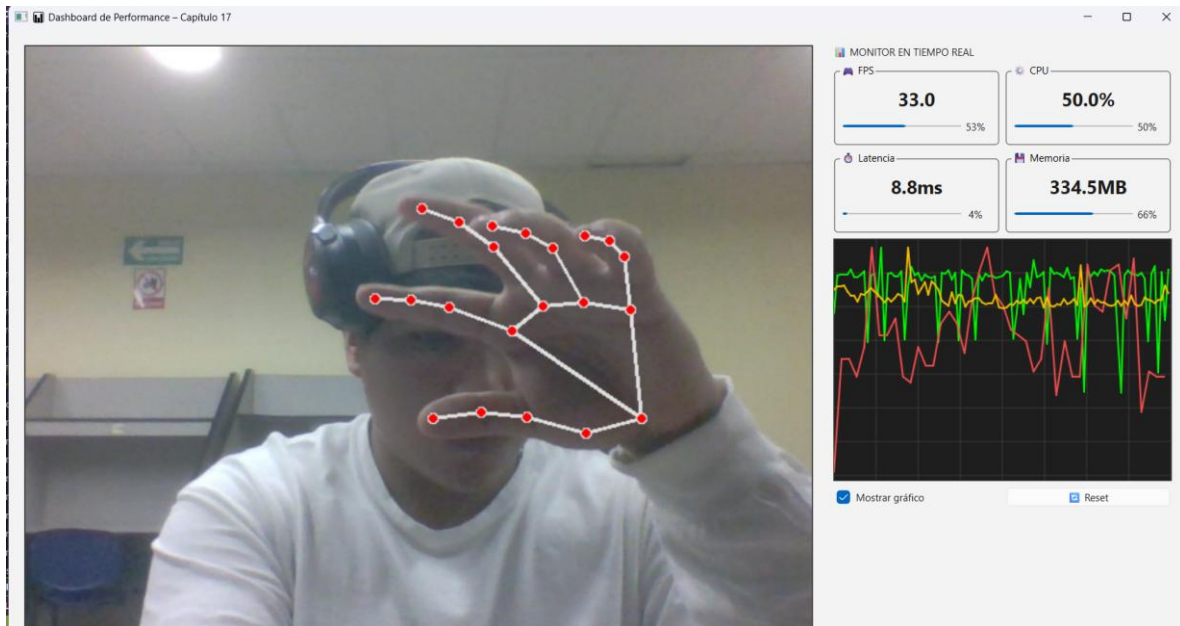
Python
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⋮

```

PS C:\Users\1171a\Downloads\Ejercicios_PV & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:/Users/1171a/Downloads/Ejercicios_PV/ejercicio16.py
cv2.circle(frame,(0['x'],0['y']),0['radio'],COLORES[0['tipo']],1)
cv2.error: OpenCV(4.13.0) :-1: error: (-5:Bad argument) in function 'circle'
> Overload resolution failed:
> - Can't parse 'center'. Sequence item with index 1 has a wrong type
> - Can't parse 'center'. Sequence item with index 1 has a wrong type
PS C:\Users\1171a\Downloads\Ejercicios_PV

```

## Ejercicio 17



```
ejercicio17.py > ...
9  from collections import deque
10 from PyQt6.QtWidgets import (QApplication, QMainWindow, QWidget, QVBoxLayout,
11                               QHBoxLayout, QLabel, QPushButton, QGroupBox,
12                               QGridLayout, QProgressBar, QCheckBox, QSpinBox,
13                               QTabWidget)
14 from PyQt6.QtCore import Qt, QTimer, QThread, pyqtSignal
15 from PyQt6.QtGui import QImage, QPixmap, QPainter, QColor, QPen
16
17 # --- NUEVOS IMPORTS DE MEDIAPIPE ---
18 import mediapipe as mp
19 from mediapipe.python.solutions import hands as mp_hands
20 from mediapipe.python.solutions import drawing_utils as mp_draw
21
22 # --- Hilo de captura ---
23
24 class CapturaThread(QThread):
25     frame_listo = pyqtSignal(object)
26     def __init__(self):
27         super().__init__()
28         self.running=True
29         self.cap=cv2.VideoCapture(0)
30     def run(self):
31         while self.running:
32             ret,frame = self.cap.read()
33             if ret: self.frame_listo.emit(frame)
34             else: self.msleep(5)
35     def stop(self):
36         self.running=False
37         self.wait()
38         self.cap.release()
39
```

## Ejercicio 18

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\1171a\Downloads\Ejercicios.PY> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:\Users\1171a\Downloads\Ejercicios.PY/ejercicio18.py
2. Libro AR
3. Probador Virtual
4. AR Catcher
5. Dashboard AR
selecciona (1-5):
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

=====
📦 EMPAQUETADOR PROFESIONAL DE APPS AR
=====

Proyectos disponibles:
1. Filtros Snapchat
2. Libro AR
3. Probador Virtual
4. AR Catcher
5. Dashboard AR

Selecciona (1-5): 1
```

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

1. Filtros Snapchat
2. Libro AR
3. Probador Virtual
4. AR Catcher
5. Dashboard AR

Selecciona (1-5): 1

Modo de empaquetado:
1. Un solo archivo (.exe)
2. Carpeta (más rápido)
Elige (1/2): 1
```

```
EXPLORER
EJERCICIOS.PY
> build
> dist
() eee.json
() eeeeeeeeeee.json
ejercicio1.py
ejercicio2.py
ejercicio3.py
ejercicio4.py
ejercicio5.py
ejercicio6.py
ejercicio7.py
ejercicio8.py
ejercicio9.py
ejercicio10.py
ejercicio11.py
ejercicio12.py
ejercicio13.py
ejercicio14.py
ejercicio15.py
ejercicio16.py
ejercicio17.py
ejercicio18.py
Filtros Snapchat.spec
libro_ar_20260521_182359.png
marcador_0.png
marcador_1.png
marcador_2.png
marcador_3.png
marcador_4.png
marcador_5.png
OUTLINE
TIMELINE

ejercicio18.py X
ejercicio18.py > main
89
90
91 def main():
92     print("=" * 55)
93     print("📦 EMPAQUETADOR PROFESIONAL DE APPS AR")
94     print("=" * 55)
95
96     proyectos = {
97         "1": ("Filtros Snapchat", "ejercicio13.py"),
98         "2": ("Libro AR", "ejercicio14.py"),
99         "3": ("Probador Virtual", "ejercicio15.py"),
100        "4": ("AR Catcher", "ejercicio16.py"),
101        "5": ("Dashboard AR", "ejercicio17.py"),
102    }
103
104    print("\nProyectos disponibles:")
105    for k,(n,) in proyectos.items(): print(f" {k}. {n}")
106    opcion = input("\nSelecciona (1-5): ").strip()
107
108    if opcion not in proyectos:
109        print("❌ opción no válida."); return
110
111    nombre, script = proyectos[opcion]
112    if not os.path.exists(script):
113
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\1171a\Downloads\Ejercicios.PY> & C:\Users\1171a\AppData\Local\Programs\Python\Python311\python.exe c:\Users\1171a\Downloads\Ejercicios.PY/ejercicio18.py
yinstaller_hooks_contrib\stdhooks'
19213 INFO: Processing standard module hook 'hook-sounddevice.py' from 'C:\Users\1171a\AppData\Local\Programs\Python\Python311\Lib\site-packages\PyInstaller\hooks_contrib\stdhooks'
19302 INFO: Processing pre-safe-import-module hook 'hook-tensorflow.py' from 'C:\Users\1171a\AppData\Local\Programs\Python\Python311\Lib\site-packages\PyInstaller\hooks_contrib\pre_safe_import_module'
19585 INFO: Processing standard module hook 'hook-PyQt6.py' from 'C:\Users\1171a\AppData\Local\Programs\Python\Python311\Lib\site-packages\PyInstaller\hooks'
20151 INFO: Processing standard module hook 'hook-PyQt6.QtWidgets.py' from 'C:\Users\1171a\AppData\Local\Programs\Python\Python311\Lib\site-packages\PyInstaller\hooks'
```